A FLAT FINITELY PRESENTED GROUPOID WHOSE RELATION IS AN OPEN IMMERSION HAS AN ETALE QUOTIENT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-algebraic-stacks-morphisms/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The theorem that a flat groupoid in algebraic spaces gives an algebraic stack. Invoked at the start of the counted proof (source lines 7262-7264) as a cross-chapter pointer. Makes the quotient stack algebraic.
- The statement that a monomorphism of algebraic stacks with no automorphisms is an algebraic space and coincides with the quotient. Invoked inside the counted proof (source lines 7265-7270). Shows that the quotient stack is an algebraic space.
- The statements that the covering of the quotient is surjective flat and of finite presentation. Invoked inside the counted proof (source lines 7271-7278). Supplies flatness and finite presentation of the quotient morphism to the base.
- The criterion for checking a property of a representable morphism along a surjective flat finitely presented cover. Invoked near the end of the counted proof (source lines 7286-7292). Identifies the diagonal of the quotient with the given open immersion.

Source reference. Original reference algebraic-proposition-algebraic-stack-no-automorphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7268–7268. Navigation pointer only: the counted text refers to the Stacks item algebraic-proposition-algebraic-stack-no-automorphisms, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Date: stacks-project:stacks-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference bootstrap-lemma-covering-quotient: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7272–7272. Navigation pointer only: the counted text refers to the Stacks item bootstrap-lemma-covering-quotient, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference bootstrap-theorem-final-bootstrap: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4487–4487. Navigation pointer only: the counted text refers to the Stacks item bootstrap-theorem-final-bootstrap, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference criteria-theorem-flat-groupoid-gives-algebraic-stack: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 3850–3850. Navigation pointer only: the counted text refers to the Stacks item criteria-theorem-flat-groupoid-gives-algebraic-stack, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-descent-lemma-flat-finitely-presented-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 7277–7277. Navigation pointer only: the counted text refers to the Stacks item spaces-descent-lemma-flat-finitely-presented-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-diagonal-unramified-morphism: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4660–4660. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-diagonal-unramified-morphism, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-flat-permanence: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5451–5451. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-flat-permanence, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference spaces-morphisms-lemma-unramified-flat-lfp-etale: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 1131–1131. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-unramified-flat-lfp-etale, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-check-property-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 479–479. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-check-property-covering, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let Y be an algebraic space. Let (U, R, s, t, c) be a groupoid in algebraic spaces over Y. Assume $U \to Y$ is flat and locally of finite presentation and $R \to U \times_Y U$ an open immersion. Then $X = [U/R] = U/R$ is an algebraic space and $X \to Y$ is étale.*

Proof. The quotient stack $[U/R]$ is an algebraic stacks by Criteria for Representability, Theorem criteria-theorem-flat-groupoid-gives-algebraic-stack. On the other hand, since $R \to U \times U$ is a monomorphism, it is an algebraic space (by our abuse of language and Algebraic Stacks, Proposition algebraic-proposition-algebraic-stack-no-automorphisms) and of course it is equal to the algebraic space U/R (of Bootstrap, Theorem bootstrap-theorem-final-bootstrap). Since $U \to X$ is surjective, flat, and locally of finite presentation (Bootstrap, Lemma bootstrap-lemma-covering-quotient) we conclude that $X \to Y$ is flat and locally of finite presentation by Morphisms of Spaces, Lemma spaces-morphisms-lemma-flat-permanence and Descent on Spaces, Lemma spaces-descent-lemma-flat-finitely-presented-permanence. Finally, consider the cartesian diagram

$$\begin{array}{ccc} R & \longrightarrow & U \times_Y U \\ \downarrow & & \downarrow \\ X & \longrightarrow & X \times_Y X \end{array}$$

Since the right vertical arrow is surjective, flat, and locally of finite presentation (small detail omitted), we find that $X \to X \times_Y X$ is an open immersion as the top horizontal arrow has this property by assumption (use Properties of Stacks, Lemma stacks-properties-lemma-check-property-covering). Thus $X \to Y$ is unramified by Morphisms of Spaces, Lemma spaces-morphisms-lemma-diagonal-unramified-morphism. We conclude by Morphisms of Spaces, Lemma spaces-morphisms-lemma-unramified-flat-lfp-etale. □